



Science

Year 9

Cambridge Lower Secondary
Student Manual



Prime Books • Science Year 9 • Student Book

First edition, 2026

Published by Prime Books, the publishing imprint of Prime School, www.primeschool.pt

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British English edition.

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INTRODUCTION

How this book works

Almost every atom of carbon in your body was once carbon dioxide in the air, and photosynthesis put it there. In this first unit you will follow carbon from the atmosphere into living things and back again, and see why adding ancient carbon to that cycle is changing the climate.

What makes this book different

- **Everything is original.** Every sentence, example and diagram was made for this book.

- **Every claim is checked.** Facts come from real, cited sources; every numerical answer was executed by computer before publication.

- **Thinking is taught explicitly.** Dedicated panels show you how specialists in this subject actually reason.

A note to teachers and parents

This book follows the spirit of an international lower-secondary programme, interpreted afresh for Prime School. Each topic opens with clear outcomes and closes with self-assessment, so progress is visible to pupil and teacher alike.

HOW TO USE THIS BOOK

Reading the page

Each section follows the same rhythm, so you always know where you are.

PANEL	WHAT IT DOES
Learning outcomes	What you will be able to do by the end.
Getting started	A warm-up that wakes up what you know.
Key words	The vocabulary of the topic, defined.
Worked example	A model solution, with the reasoning shown.
Questions	Practice, from straightforward to demanding.
Investigation	A hands-on experiment with equipment.
Safety	How to carry out the work safely.
Think like a scientist	A task that builds scientific thinking.
Summary checklist	'I can' statements to check your progress.

How to work

Read with a pencil in your hand. When the book asks a question, try to answer it before reading on. Do the work in your notebook, and check your reasoning, not just your answer.

GETTING SET UP

Before Unit 1

You need a pencil, a ruler, a notebook and your curiosity. Anything else this unit requires is listed where it is needed.

Your toolkit

- A notebook for working, planning and recording, not just for final answers.
- The habit of checking: recalculate, reread, and ask whether the answer is sensible.
- The confidence to say "I am not sure yet", which is where all real learning starts.

A quick self-check

- 1 What do you already know about this topic?
- 2 Which part of it do you expect to find hardest, and why?
- 3 How will you know when you have genuinely understood something, rather than just recognised it?

Unit 1 begins now. Work carefully, check honestly, and ask good questions.

1

Photosynthesis and the carbon cycle

How life captures sunlight, and why the climate is changing

© IN THIS UNIT, YOU WILL

- ✓ Explain photosynthesis and write its equation.
- ✓ Describe how leaves are adapted to their job.
- ✓ Investigate the factors that limit photosynthesis.
- ✓ Trace carbon through the carbon cycle.
- ✓ Explain the evidence for climate change.

◆ GETTING STARTED

Almost every atom of carbon in your body was once carbon dioxide in the air. In pairs, discuss:

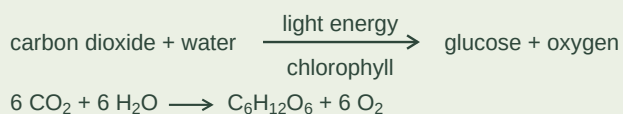
- how do you think carbon got from the air into your body?
- what living process could possibly have started that journey?
- where might those carbon atoms go after you have finished with them?

UNIT 1 • TOPIC 1.1

Photosynthesis

Learn

Photosynthesis is the process that keeps almost every living thing alive. In it, plants use light energy to build glucose from carbon dioxide and water. It is the only significant way that energy enters the living world.



Photosynthesis happens inside **chloroplasts**, which contain the green pigment **chlorophyll**. Chlorophyll absorbs light, mostly red and blue light, and reflects green, which is why leaves look green to you.

Notice the elegance of it: respiration, which you met in Year 8, is almost exactly the reverse. Photosynthesis stores energy; respiration releases it.



Watch: Photosynthesis

More about photosynthesis

When the sun is bright, leaves often build more sugar than they can spend. Rather than waste it, the plant banks the extra, changing it into starch and tucking it away inside its chloroplasts and other cells. Later, on a dull day or during the night, the starch is turned back into sugar and used. A simple starch test on a leaf tells you whether photosynthesis has been happening there.

And because only some plant cells own chloroplasts, only those cells can photosynthesise. The glucose made by photosynthesis has three main uses. It is used in respiration, releasing energy for growth. It is converted into starch for storage. And it is turned into cellulose for cell walls, and into proteins for growth, when combined with minerals from the soil.



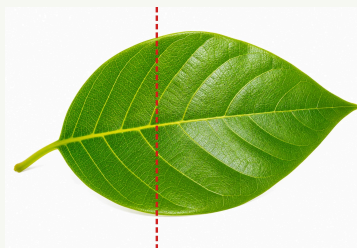
Figure 1.1 A chloroplast, where light energy is captured. Photosynthesis happens inside the chloroplasts, which are inside some of the cells in a leaf.

Chloroplasts and chlorophyll

Chloroplasts are small green organelles in plant cells. A single leaf cell may contain fifty or more. Each has two membranes and is filled with fluid called the stroma. Suspended in it are stacks of flattened discs called grana; each disc is a thylakoid membrane, and this is where the chlorophyll sits. Chlorophyll is a pigment: it strongly absorbs red and blue light and uses their energy for photosynthesis, but it cannot use green light, so green is reflected into your eyes. That is why plants look green.

A look inside a leaf

Chloroplasts are mostly inside the cells in the middle layer of the leaf. Leaves are very thin, so it is easy for sunlight to reach these cells. Chloroplasts also need lots of water, which comes to the leaf through the veins. Carbon dioxide diffuses into the leaf from the air through tiny holes called stomata. In singular form, one hole is called a stoma. Oxygen also diffuses out through the air spaces after photosynthesis and respiration. A leaf is so thin that, if you cut one across and place the cut edge under a microscope, all of these layers can be seen clearly, as Figure 1.2 shows.



A leaf, with its veins bringing water to every cell. The red dashed line shows where to cut to see the layers drawn in Figure 1.2.

Leaf structure and function

PART	FUNCTION
waxy cuticle	A waterproof layer on the surface. It reduces water loss without blocking light.
upper epidermis	A thin, transparent layer of cells. It lets light pass straight through to the cells below.
palisade layer	Tall, tightly packed cells stuffed with chloroplasts. This is where most photosynthesis happens.
spongy mesophyll	Rounded cells with air spaces between them, letting carbon dioxide diffuse to every cell.
vein (xylem and phloem)	The leaf's plumbing. Xylem brings water up from the roots; phloem carries the glucose away.
lower epidermis with stomata	The underside is dotted with stomata, each guarded by two guard cells, letting gases in and out.

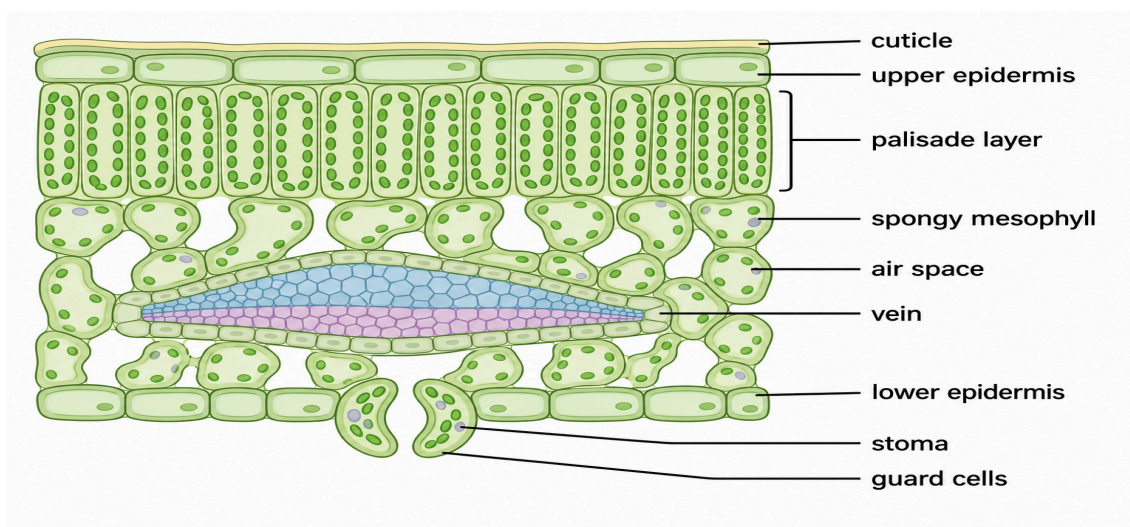


Figure 1.2 A cross-section through a leaf.

QUESTIONS

- 1 Write the word equation and the balanced symbol equation for photosynthesis.
- 2 Where in a plant cell does photosynthesis happen?
- 3 Why do leaves appear green?
- 4 Compare photosynthesis with respiration. Give one similarity and two differences.
- 5 A plant is kept in complete darkness for a week. Explain what happens to it and why.

Key words to remember

chlorophyll	the green pigment that captures light
stoma	a tiny hole guarded by guard cells
xylem	vein tissue carrying water from roots

· ACTIVITY 1: WHICH SURFACE OF A LEAF HAS MOST STOMATA?

Work with a partner. Paint clear nail varnish onto the upper surface of a leaf and, separately, onto the lower surface. Let both films dry, peel each one off the leaf, and look at them under a microscope.

Count the stomata you can see in each film. Repeat with three leaves and record your counts in a table. Then answer: which surface has more stomata, and why might that be an advantage to the plant?

UNIT 1 · TOPIC 1.2

Factors limiting photosynthesis

Learn

A leaf is a solar panel with plumbing. Every feature has a purpose.

FEATURE	HOW IT HELPS
broad and flat	A large surface area to catch light.
thin	Gases diffuse only a short distance.
many chloroplasts near the top	Where the light is strongest.
stomata on the underside	Let carbon dioxide in and oxygen out.
veins	Bring water in and carry glucose away.
waxy cuticle	Reduces water loss without blocking light.

Photosynthesis does not always run at full speed. A limiting factor is whatever is holding it back at that moment: usually light intensity, carbon dioxide concentration or temperature. If you increase light intensity, the rate rises, then levels off, because something else has become the limiting factor.

Temperature behaves differently: the rate rises, peaks, then falls sharply, because the enzymes that drive photosynthesis are damaged, or denatured, by heat.

KEYWORDS

photosynthesis: making glucose from carbon dioxide and water using light.

chlorophyll: the green pigment that absorbs light energy.

chloroplast: the part of a plant cell where photosynthesis happens.

stoma: a tiny pore in a leaf that lets gases in and out. (Plural: stomata.)

limiting factor: the factor that is currently restricting the rate.

You will need: pondweed (*Elodea*), a test tube, a beaker of water, a lamp, a ruler, sodium hydrogen carbonate solution, a stopwatch.

Method:

- 1 Place the pondweed in the test tube of sodium hydrogen carbonate solution, which supplies carbon dioxide.
- 2 Stand the tube in a beaker of water. This acts as a heat shield, so the lamp warms the plant as little as possible.
- 3 Place the lamp 10 cm away. Wait two minutes for the rate to settle.
- 4 Count the oxygen bubbles released in one minute. Repeat twice and take a mean.
- 5 Repeat at 20 cm, 30 cm, 40 cm and 50 cm.
- 6 Plot a graph of bubbles per minute against distance.

SAFETY

The lamp becomes hot. Do not touch the bulb and keep water away from the electrical fitting. Wash your hands after handling pond water.

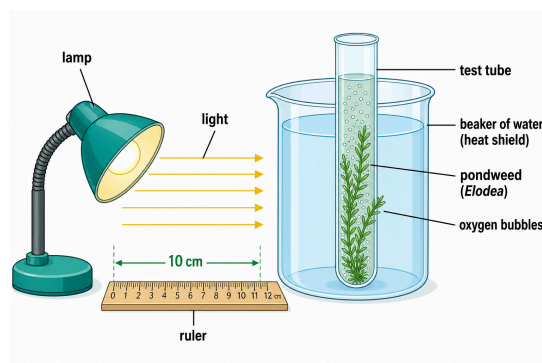


Figure 1.3 Light intensity affects photosynthesis from different distances.

∴ THINK LIKE A SCIENTIST

In this investigation, why is the beaker of water essential? Which variable would become confounded without it, and how would that affect your conclusion?

• QUESTIONS

- 1 Give three adaptations of a leaf and explain how each helps photosynthesis.
- 2 What is a limiting factor?
- 3 Sketch the graph of rate against light intensity. Explain the shape of each part.
- 4 Explain why the rate of photosynthesis falls at very high temperatures, using the word denatured.
- 5 In the investigation, why was a mean of three counts taken?

Minerals and plant growth

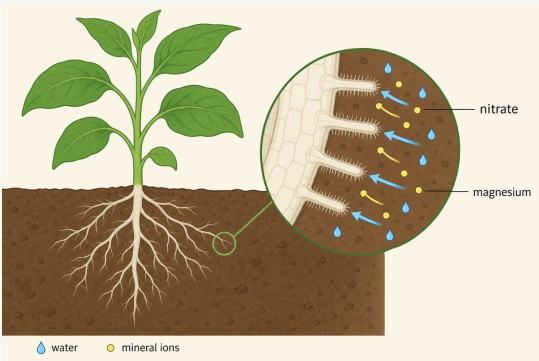
Learn

Photosynthesis builds sugar from gas and water, but sugar is not all a plant needs. To make **proteins**, a plant must have nitrogen. To make **chlorophyll**, it must have magnesium. These raw materials arrive not as food but as **minerals**: salts dissolved in the water that roots draw up from the soil.

Roots are built for collecting. Fine hairs just behind the growing tip multiply the surface area, and each hair takes in water and dissolved minerals. Two minerals matter most. **Nitrates** supply nitrogen for growth: without them a plant is stunted and its leaves turn pale yellow, because it cannot build the proteins for new cells. **Magnesium ions** are needed to make chlorophyll: a shortage shows as yellowing leaves, because without magnesium the green pigment cannot be made and photosynthesis slows.

This is why farmers and gardeners add **fertiliser**. A fertiliser replaces the minerals that crops remove from the soil, so the next crop can grow as fast as the last. But more is not always better. Too much fertiliser drains into streams and ponds, where it feeds algae instead, which then bloom, die and rot, using up the oxygen that fish need.

Figure 1.4 How roots take up minerals. Water and dissolved mineral salts enter the fine root hairs just behind the growing tip. Nitrates supply nitrogen for proteins; magnesium ions are needed to make chlorophyll.



Minerals and what they do

MINERAL	PROVIDES	SIGNS OF SHORTAGE
Nitrate	Nitrogen, for proteins and growth	Poor growth; small, pale yellow leaves

Minerals and plant growth

· QUESTIONS

1 A farmer's wheat grows slowly and its leaves are pale yellow. Which mineral is the crop short of, and why does the shortage show in the leaves?

2 Explain why magnesium shortage slows photosynthesis as well as growth.

3 Root hairs take up minerals dissolved in water. Explain how their shape helps them do this.

4 Give one benefit and one drawback of adding fertiliser to a field.

The carbon cycle

Learn

Carbon atoms are never created or destroyed. They circulate endlessly between the air, living things, the oceans and the rocks. This circulation is the **carbon cycle**.

Carbon goes OUT of the air by:

- **photosynthesis**, when plants build glucose from carbon dioxide

Carbon returns TO the air by:

- **respiration**, in plants, animals and microbes
- **decomposition**, when microbes break down dead material
- **combustion**, when fuels are burned



Watch: The carbon cycle

For millions of years these processes roughly balanced. Some carbon, though, was locked away underground as **fossil fuels**: coal, oil and natural gas, formed from organisms that died long ago and did not fully decompose. That carbon was out of the cycle entirely.

When we burn fossil fuels, we return that ancient carbon to the air in a few centuries, far faster than photosynthesis and the oceans can remove it. The cycle no longer balances, and carbon dioxide accumulates.



Figure 1.2 The carbon cycle, showing the routes carbon takes.

• QUESTIONS

- 1 Name the process that removes carbon dioxide from the air.
- 2 Name three processes that return carbon dioxide to the air.
- 3 Explain how fossil fuels formed, and why burning them changes the cycle.
- 4 A forest is cut down and burned. Give two reasons why this raises atmospheric carbon dioxide.
- 5 Explain why planting trees can help, but cannot by itself solve the problem.

✦ DID YOU KNOW?

Carbon is a champion recycler. A carbon atom in your breath this morning may have been in a dinosaur, a limestone cliff, a fern in a coal seam and an ocean wave, and it will still be circulating long after you have gone.

Climate change

Learn

Some gases in the atmosphere, called **greenhouse gases**, let sunlight through but absorb the infrared radiation the Earth emits. This keeps the planet warm enough for life, and is called the **greenhouse effect**. It is entirely natural and entirely necessary.

The problem is its strengthening. By adding carbon dioxide and methane to the atmosphere, human activity is trapping more heat than before. This is **enhanced global warming**, and the resulting shifts in weather patterns are **climate change**.

The evidence

The measurements are real, public and easy to check. At the Mauna Loa Observatory in Hawaii, carbon dioxide has been recorded continuously since 1958.

YEAR	ATMOSPHERIC CO ₂ (ppm)
pre-industrial (before ~1750)	about 280
1958 (record begins)	315
2000	369
2015	402
2025	429.6
June 2026	431.4

That is a rise of more than 50% above the pre-industrial value, and the rate of increase is itself increasing. Ice cores show that for the previous 800 000 years, carbon dioxide never rose above about 300 ppm.

Consequences

- **Rising global mean temperature**, and more frequent heatwaves.
- **Melting ice sheets and glaciers**, and **rising sea level**.
- **Ocean acidification**, as seawater absorbs carbon dioxide.
- **Shifting rainfall**, bringing drought to some regions and floods to others.

Portugal is directly affected. Longer, hotter summers raise wildfire risk in the interior, and the Algarve faces increasing pressure on its water supply.

∴ THINK LIKE A SCIENTIST

Scientists do not rely on a single measurement. The Mauna Loa record is supported by ice cores, satellite data, ocean buoys and thousands of independent stations, all agreeing. Explain why this **convergence of independent evidence** makes a conclusion far stronger than any one dataset could.

• QUESTIONS

- 1 Explain the natural greenhouse effect, and why life needs it.
- 2 Using the table, calculate the rise in carbon dioxide from 1958 to June 2026.
- 3 What percentage increase is 431.4 ppm compared with the pre-industrial 280 ppm?
Show your working.
- 4 Give three consequences of enhanced global warming.
- 5 Describe two effects of climate change that matter specifically in Portugal.

☒ SUMMARY CHECKLIST

- ✓ I can write the equation for photosynthesis and explain it.
- ✓ I can explain how leaves are adapted to photosynthesis.
- ✓ I can identify and explain limiting factors.
- ✓ I can trace carbon through the carbon cycle.
- ✓ I can describe the evidence for climate change and evaluate it.

UNIT 1 • FINAL PROJECT

A carbon audit of your school

★ FINAL PROJECT

A carbon audit of your school

Work in groups to investigate where carbon dioxide enters and leaves your school grounds, then propose one realistic change that would reduce net emissions. Present your findings as a scientific poster with data.

What to hand in

- A labelled map of your school showing carbon sources and carbon sinks.
- At least three real measurements or counts you made yourselves.
- One realistic proposal, with an estimate of its effect and your reasoning.
- A sources list, with links, for any external data you used.

How it will be judged

| Criterion | Weight | |---|---| | Scientific accuracy of the carbon analysis | 35% | | Quality and honesty of your own data | 30% | | Realism of the proposal and its justification | 25% | | Presentation and referencing | 10% |

UNIT 1 • CHECK YOUR PROGRESS

- 1 Write the balanced symbol equation for photosynthesis. [2]
- 2 Name the pigment that absorbs light, and the structure that contains it. [2]
- 3 Give three leaf adaptations and explain each. [3]
- 4 Explain what is meant by a limiting factor, with an example. [2]
- 5 Name one process that removes carbon dioxide from the air and three that return it. [4]
- 6 Explain why burning fossil fuels unbalances the carbon cycle. [2]
- 7 The Mauna Loa record shows 315 ppm in 1958 and 431.4 ppm in June 2026.
a. Calculate the increase. b. Calculate the mean annual rise. [2]
- 8 Explain why the greenhouse effect is natural but enhanced global warming is a problem. [2]

UNIT 1 • EVALUATION

Be honest. Colour the circle that matches how you feel about each outcome: green means confident, amber means nearly there, red means needs more work.

I CAN ...	GREEN	AMBER	RED
explain photosynthesis and its equation	☐	☐	☐
explain leaf adaptations	☐	☐	☐
investigate limiting factors	☐	☐	☐
trace carbon through the carbon cycle	☐	☐	☐
evaluate the evidence for climate change	☐	☐	☐

UNIT 1 • WHAT CAN YOU DO?

✓ WHAT CAN YOU DO?

- | | |
|---|---|
| ☐ Write and explain the equation for photosynthesis. | ☐ Trace carbon through the carbon cycle. |
| ☐ Explain how a leaf is adapted to its function. | ☐ Describe and evaluate the evidence for climate change. |
| ☐ Identify limiting factors and interpret rate graphs. | |

■ UNIT 1 ROUND-UP

photosynthesis · chlorophyll · chloroplast · stoma · limiting factor · denatured · carbon cycle · decomposition · combustion · fossil fuel · greenhouse gas · greenhouse effect · climate change · ocean acidification

GLOSSARY

Unit 1

photosynthesis: making glucose from carbon dioxide and water using light.

chlorophyll: the green pigment that absorbs light energy.

chloroplast: the part of a plant cell where photosynthesis happens.

stoma: a tiny pore in a leaf that lets gases in and out.

limiting factor: the factor currently restricting the rate.

denatured: changed in shape by heat so it no longer works.

carbon cycle: the circulation of carbon between air, life, oceans and rocks.

greenhouse effect: the trapping of infrared radiation by atmospheric gases.

ocean acidification: the fall in ocean pH as seawater absorbs carbon dioxide.

mineral: a salt dissolved in soil water that a plant needs, such as nitrate or magnesium.

fertiliser: a substance added to soil to replace minerals that crops have used up.

SOURCES & REFERENCES

Where everything came from

All text and diagrams are original to this edition. The factual claims can be checked on page 20. Links were live as of July 2026.

§ UNIT 1 • PHOTOSYNTHESIS AND THE CARBON CYCLE

- 1 **Atmospheric carbon dioxide at Mauna Loa. The monthly mean for June 2026 was 431.44 ppm; the pre-industrial value was about 280 ppm.**
<https://gml.noaa.gov/ccgg/trends/>
- 2 **The 2025 seasonal peak. Atmospheric carbon dioxide peaked at 430.5 ppm in May 2025.**
<https://gml.noaa.gov/ccgg/trends/weekly.html>
- 3 **Ice-core evidence. For the previous 800 000 years, carbon dioxide never exceeded about 300 ppm.**
<https://www.climate.gov/news-features/understanding-climate/climate-change-atmospheric-carbon-dioxide>
- 4 **Photosynthesis. The light-driven reaction that builds glucose from carbon dioxide and water.**
<https://en.wikipedia.org/wiki/Photosynthesis>



UNIT 2

Properties of materials

What everything is made of, and why it behaves as it does

IN THIS UNIT, YOU WILL

- Describe atoms, protons, neutrons and electrons
- Read the Periodic Table and its trends
- Explain why elements react to form compounds
- Distinguish simple and giant structures
- Use symbols and formulae confidently

Atomic structure and the Periodic Table

Learn

Everything around you is built from about one hundred different **elements**. Each element is made of tiny particles called atoms. An atom is not the smallest thing there is: inside it are even smaller particles, called **protons**, **neutrons** and **electrons**.

Protons and neutrons sit together in the **nucleus** at the centre of the atom. Protons carry a positive charge and neutrons carry no charge at all. Electrons, which are negative, move around the nucleus in shells. In a neutral atom the number of electrons equals the number of protons, so the charges cancel.

Two numbers describe any atom. The **atomic number** counts the protons; it fixes the element's identity and its place in the Periodic Table. The **mass number** counts protons plus neutrons. A lithium atom, for example, has atomic number 3 and mass number 7, so it has 3 protons, 3 electrons and 4 neutrons.

The **Periodic Table** lists the elements in order of atomic number. Rows are called periods and columns are called groups. Elements in the same group have the same number of electrons in their outer shell, which is why they behave in similar ways.

Calculating protons, neutrons and electrons

The rules. The atomic number (proton number) is the number of protons. The mass number is protons + neutrons. In a neutral atom, electrons = protons. So:

protons = atomic number · neutrons = mass number – atomic number · electrons = atomic number

Worked example: chlorine, atomic number 17, mass number 35

STEP	CALCULATION	HOW YOU WRITE IT
Protons	= atomic number	= 17 protons
Neutrons	= mass number – atomic number = 35 – 17	= 18 neutrons
Electrons	= protons (neutral atom)	= 17 electrons

Try one yourself. Sodium has atomic number 11 and mass number 23. Work out the protons, neutrons and electrons, then

· QUESTIONS

- 1 An atom has 11 protons and mass number 23. Name the element and work out how many neutrons and electrons it has.
- 2 Why do sodium and potassium, which sit in the same group, react in similar ways?
- 3 Explain why an atom has no overall charge.

Inside the atom

Inside an atom. Almost all of the mass sits in the tiny nucleus at the centre, where protons, marked +, and neutrons cluster. Electrons, marked −, move around the nucleus in shells. In a neutral atom the number of electrons equals the number of protons, so the charges cancel exactly. The atomic number counts the protons and fixes the element's place in the Periodic Table, shown as a fragment in the corner: each row is a period, each column a group. Notice the carbon atom drawn here: 6 protons and 6 electrons, with 2 electrons in the inner shell and 4 in the outer one. That outer shell of four is what lets carbon bond with up to four other atoms, which is why carbon forms so many different compounds.

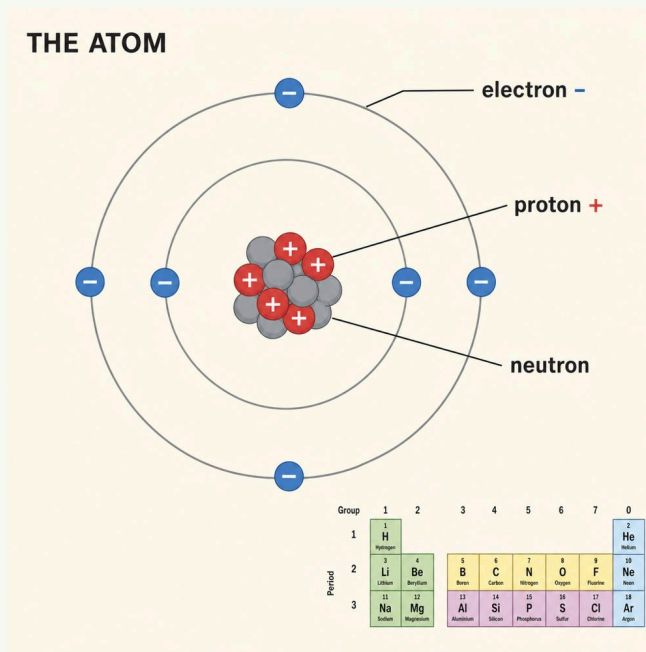


Figure 2.1

· ACTIVITY

Build a model atom. Choose any element from the first twenty of the Periodic Table. Draw its atom, showing the nucleus with the correct numbers of protons and neutrons, and the electrons arranged in shells. Label the atomic number and the mass number. Then write two sentences: one on why the atom has no overall charge, and one on what makes your element unique in its group.

Partner challenge. Swap drawings with a partner. From their diagram alone, work out which element they chose. Check your answer against the Periodic Table.

Trends in groups within the Periodic Table

Learn

A quick look along a group reveals patterns that chemists rely on. **Group 1** contains lithium, sodium and potassium: soft metals that float on water and react vigorously with it, producing hydrogen gas. As you go down the group the metals become more reactive, because the outer electron is further from the nucleus and more easily lost.

The **halogens** in Group 7 show the opposite journey: fluorine and chlorine are reactive gases, bromine is a liquid, and iodine is a dark solid. Reactivity falls as you descend, because the incoming electron is further from the nucleus and less strongly attracted.

The **noble gases** in Group 0 are the odd ones out. Their outer shells are full, so they neither lose nor gain electrons. That makes them almost completely unreactive, which is why they glow steadily in lamps and protect hot filaments from burning.

PERIODIC TABLE OF THE ELEMENTS

Legend:

- 1 Alkali Metals (Pink)
- 2 Alkaline Earth Metals (Orange)
- 3-12 Transition Metals (Yellow)
- 13 Post-Transition Metals (Green)
- 14 Metalloids (Teal)
- 15 Other Nonmetals (Light Blue)
- 16 Chalcogens (Purple)
- 17 Halogens (Pink)
- 18 Noble Gases (Light Blue)

Elements shown include: H, He, Li, Be, Na, Mg, K, Ca, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Ga, Ge, As, Se, Br, Kr, Rb, Sr, Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd, In, Sn, Sb, Te, I, Xe, Cs, Ba, La, Hf, Ta, W, Re, Os, Ir, Pt, Au, Hg, Tl, Pb, Bi, Po, At, Rn, Fr, Ra, Ac, Th, Pa, U, Np, Pu, Am, Cm, Bk, Cf, Es, Fm, Md, No, Lr, and various lanthanides and actinides.

Figure 2.3 The Periodic Table: Group 1 (alkali metals), Group 7 (halogens) and Group 0 (noble gases) are colour-coded.

QUESTIONS

- 1 Sodium and chlorine react vigorously together. Use ideas about outer electrons to explain why.
- 2 Predict whether astatine, below iodine in Group 7, is a solid, liquid or gas at room temperature, and give a reason.
- 3 Give two uses of noble gases and explain why each use depends on their unreactivity.

The metals and the non-metals: their trends

The trends for the reactive metals of Groups 1 and 2, then separately for the non-metals of Groups 6 and 7.

Group 1 · alkali metals

ELEMENT	ATOMIC	MASS	ELECTRONIC	MELT (°C)	BOIL (°C)
Lithium	3	7	2,1	181	1347
Sodium	11	23	2,8,1	98	883
Potassium	19	39	2,8,8,1	63	774

- Melting points fall down the group: the outer electron is lost more easily, so the atoms are held less tightly.
- Reactivity rises down the group: lithium fizzes gently on water, sodium skates furiously across it, and potassium bursts into lilac flame.

Group 2 · alkaline earth metals

ELEMENT	ATOMIC	MASS	ELECTRONIC	MELT (°C)	BOIL (°C)
Beryllium	4	9	2,2	1287	2470
Magnesium	12	24	2,8,2	650	1090
Calcium	20	40	2,8,8,2	839	1487

- Melting points are high but irregular down the group.
- Reactivity also rises down the group, though these metals are harder and less violent with water than their Group 1 neighbours.

Group 6 · chalcogens

ELEMENT	ATOMIC	MASS	ELECTRONIC	MELT (°C)	BOIL (°C)
Oxygen	8	16	2,6	-219	-183
Sulfur	16	32	2,8,6	113	445

- Melting and boiling points rise steeply down the group, from a gas to a solid.
- Reactivity falls down the group: oxygen grabs electrons more fiercely than sulfur does.

Group 7 · halogens

ELEMENT	ATOMIC	MASS	ELECTRONIC	MELT (°C)	BOIL (°C)
Fluorine	9	19	2,7	-220	-188
Chlorine	17	35	2,8,7	-101	-35

- Melting and boiling points rise down the group: the molecules grow heavier and attract each other more.
- Reactivity falls down the group: fluorine is the fiercest of all, while iodine is the gentlest of the halogens.

Materials of Groups 1, 2 and 7

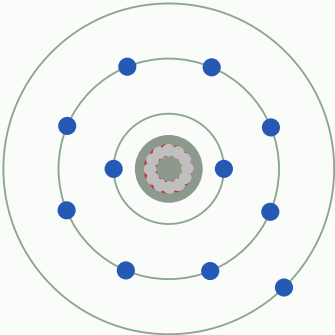
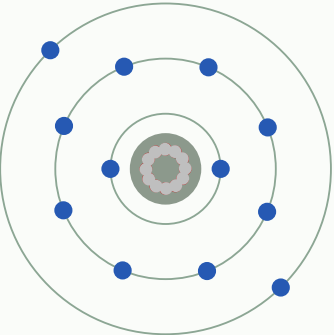
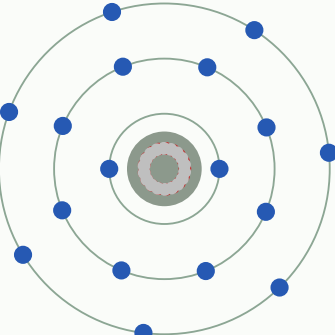
Group 1 · alkali metals. These are soft enough to cut with a knife, light enough to float on water, and silvery when freshly cut, though the shine soon tarnishes in air. They are kept under oil, because both air and moisture attack them. Lithium, sodium and potassium all melt below 200 °C, so a warm day leaves them soft and sluggish. Their low density and eagerness to give away an outer electron make them useful in batteries and in street lamps that glow orange.

Group 2 · alkaline earth metals. Harder, denser and calmer than their Group 1 neighbours. Magnesium ribbon burns with a brilliant white light, once used in camera flashes and still used in fireworks; calcium fizzes gently in water and builds our bones and teeth along with carbon and oxygen; beryllium is light and stiff enough for specialist instruments. They hold their two outer electrons more tightly, so they need more persuasion to react.

Group 7 · halogens. Colourful, pungent non-metals. Fluorine is a pale yellow gas so reactive it attacks glass; chlorine is a greenish gas used in tiny amounts to kill bacteria in drinking water and swimming pools; bromine is one of only two elements that are liquid at room temperature, a deep red-brown, fuming liquid; iodine forms shiny grey crystals that vaporise into a violet gas. Their colours darken steadily down the group, from pale yellow to nearly black.

Electronic structures: one element from each group

One worked example each from Group 1, Group 2 and Group 7, drawn shell by shell.

SODIUM · Group 1  Na · 2,8,1 Sodium has a single outer electron, the signature of Group 1, and hands it away eagerly in every reaction.	MAGNESIUM · Group 2  Mg · 2,8,2 Magnesium has two outer electrons, the mark of Group 2, so it must give away two and works a little harder at reacting.	CHLORINE · Group 7  Cl · 2,8,7 Chlorine is one electron short of a full shell, the badge of Group 7, so it pulls one in and becomes a chloride ion, Cl⁻.
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WHAT TO NOTICE

In every case the outer shell decides behaviour: one electron to lose, two to lose, or one to gain. The group number tells you which. Down any group the shells pile up, the outer electron sits further from the nucleus, and metals lose it more easily while non-metals attract one less strongly.

Why reactivity changes down a group

Figure 2.4 Comparing the electronic structures of lithium, sodium and potassium. Each atom has a single electron in its outer shell, but look how much further that electron sits from the nucleus each time a new shell is added. The pull of the positive nucleus weakens with distance, and the inner shells shield it too, so the outer electron is easier and easier to lose.

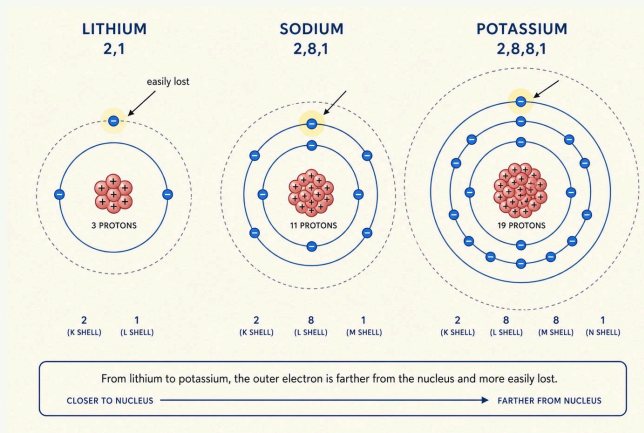


Figure 2.4

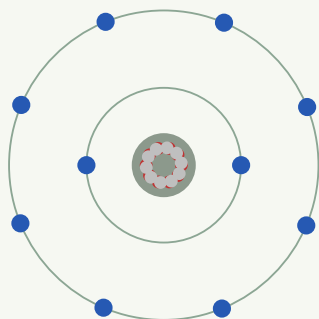
The metals of Groups 1 and 2: reactivity rises. A metal reacts by giving away its outer electrons. Down the group, each new period adds a whole shell, so the outer electron is further from the nucleus, and the full shells in between block, or shield, the nucleus's pull. A distant, shielded electron is easily lost, and an atom that loses electrons easily reacts fiercely. That is why potassium, with four shells, throws its outer electron at water more violently than sodium, which in turn beats lithium with three shells. Group 2 behaves the same way, but each atom must lose two electrons, not one, so the metals are less violent than their Group 1 neighbours.

The halogens of Group 7: reactivity falls. A halogen reacts by pulling in one extra electron to complete its shell. Down the group, the incoming electron must land in a shell further from the nucleus, where the attraction is weaker and more shielded. Fluorine's new electron drops into the second shell, close to a strong pull, so fluorine is the most reactive non-metal of all; iodine's must reach the fifth shell, far out, so iodine reacts slowly and gently. The same distance rule decides both groups, but in opposite directions.

The noble gases: why they are noble

A full shell changes everything

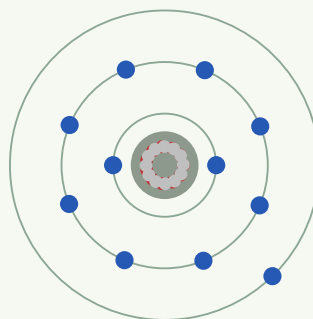
NEON · Group 0



Ne · 2,8

Nothing to give, nothing to gain. Neon's outer shell is already full, so it is content.

SODIUM · Group 1



Na · 2,8,1

That single outer electron is loose, easy to lose, and the reason sodium reacts.

Why noble? The name came from chemistry's early days, when gases that refused to react with anything were called noble, aloof and superior, standing apart from the crowd like nobility. The reason is simple and beautiful: their outer electron shells are completely full. Helium holds 2, neon 8, argon 8. An atom reacts only to complete its outer shell, by giving electrons away or pulling them in. A noble gas needs to do neither. It is born satisfied, so it has nothing to gain from reacting, and nothing to spare.

Proof by extremes. Fluorine, the hungriest element of all, attacks almost every metal, bursts water into flame, even etches glass. Yet even fluorine cannot persuade neon or argon to react under normal conditions. When even the fiercest reagent gives up, the refusal is absolute.

Useful because unreactive. Their indifference is exactly what makes them useful. Argon fills light bulbs and welding shields, because it will not burn the filament or react with hot metal. Helium lifts party balloons and airships, and its full, light atoms escape gravity's grip only slowly. Neon glows orange-red in signs when excited by electricity, then settles back, unchanged.

Why elements react to form compounds

Learn

Atoms are most stable when their outer shell of electrons is full. Atoms that lack a full outer shell react with other atoms to complete it, and the result is a **compound**: a substance in which two or more elements are chemically joined.

There are two main ways to finish an outer shell. A metal atom may **donate** its outer electrons to a non-metal, leaving a positive ion attracted to the negative ions that receive them. Sodium chloride, common salt, is built this way from Na^+ and Cl^- ions. Or two non-metal atoms may **share** electrons, as in water and carbon dioxide. Shared pairs of electrons are called **covalent bonds**.

A compound is very different from a mixture. In a mixture the ingredients keep their own properties and can be separated by physical means. In a compound the elements are chemically bonded, have new properties entirely, and can only be separated by a chemical reaction. Table salt bears no resemblance to a soft grey metal and a poisonous green gas.

Figure 2.3 A compound (left) bonds atoms into identical molecules; a mixture (right) keeps its ingredients separate.

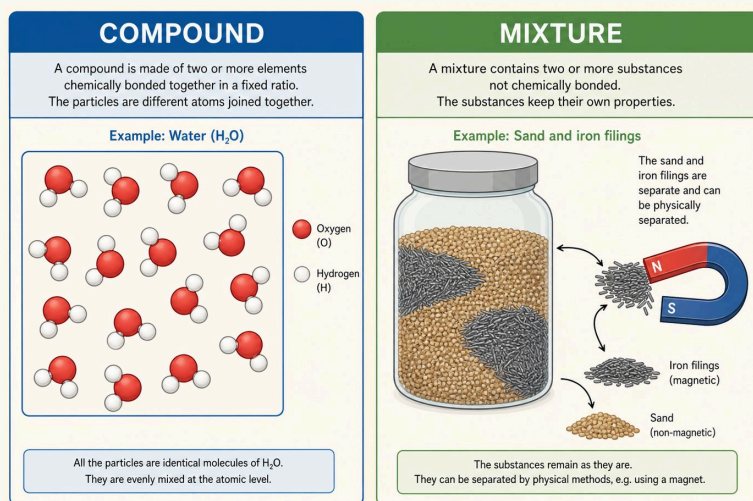


Figure 2.3 A compound compared with a mixture.

• QUESTIONS

- 1 Use ideas about electrons to explain why sodium and chlorine react to form sodium chloride.
- 2 State two differences between a compound and a mixture, with an example of each.
- 3 Water has the formula H_2O . Explain what the formula tells you about a molecule of water.

Compounds all around us

Molecules and lattices. On the left, a water molecule: one oxygen atom shares electrons with two hydrogen atoms, held by covalent bonds. Beside it, carbon dioxide: a carbon atom double-bonded to two oxygens in a straight line. On the right, sodium chloride: the sodium atoms have donated their outer electrons to the chlorine atoms, forming positive sodium ions and negative chloride ions that lock together in a regular repeating giant lattice. Look at the colour key: red is oxygen, white is hydrogen, black is carbon. Each Na^+ ion is surrounded by six Cl^- ions, and each Cl^- ion by six Na^+ ions, in every direction, which is why a grain of salt forms such perfect little cubes.

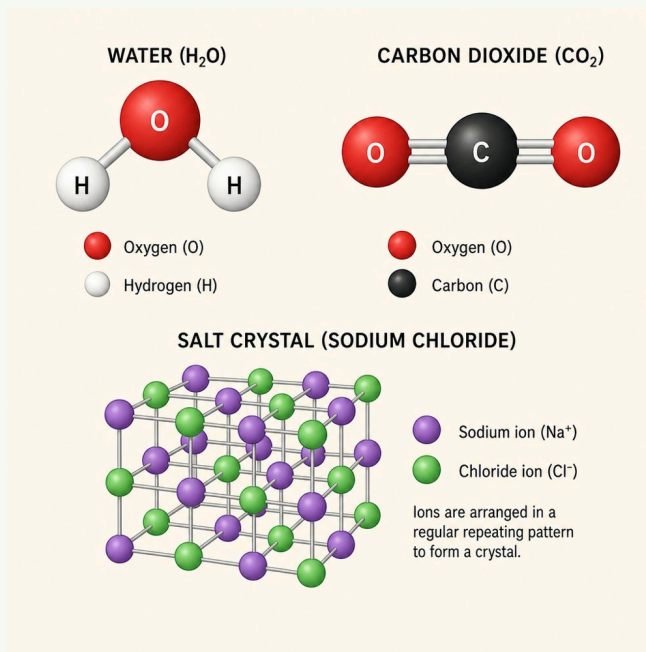


Figure 2.2

• ACTIVITY

Formula detective. Write the formulae for these compounds: sodium chloride, water, carbon dioxide, methane. For each one, state the elements present and the number of atoms of each in one unit of the compound.

Compound or mixture? Sort these into two columns, compound or mixture: sea water, table salt, air, pure water, brass, sugar. Give a one-sentence reason for each.

Model building. With a partner, use modelling clay or beads in different colours to build one molecule of water and part of the sodium chloride lattice. Photograph or sketch your models and label the bonds.

Ionic and covalent bonds

An ionic bond forms when electrons are transferred. A sodium atom has one lonely electron in its outer shell; a chlorine atom has seven and needs one more. Sodium hands its spare electron across, and in that single exchange both atoms end up with full outer shells. The sodium atom, having lost a negative electron, is left as a positive sodium ion, Na^+ ; the chlorine atom, having gained one, becomes a negative chloride ion, Cl^- . Opposite charges attract, and that attraction, holding millions of ions together in a lattice, is the ionic bond. Ionic bonds are strong, which is why salt crystals are hard and only melt at very high temperatures.

A covalent bond forms when electrons are shared. Two hydrogen atoms each hold a single electron, half a shell each. Instead of transferring, they overlap their shells and share the pair, so each atom enjoys a full shell part of the time. That shared pair is the covalent bond, drawn as a dash, $\text{H}-\text{H}$. Covalent bonding is the rule between non-metals: water, carbon dioxide and methane are all built from shared electrons. The bonds inside each molecule are strong, but the molecules attract each other only weakly, so simple molecular substances melt and boil at low temperatures.

How to tell them apart. Metal plus non-metal usually means transfer, ions and an ionic bond; two non-metals usually means sharing, molecules and covalent bonds.

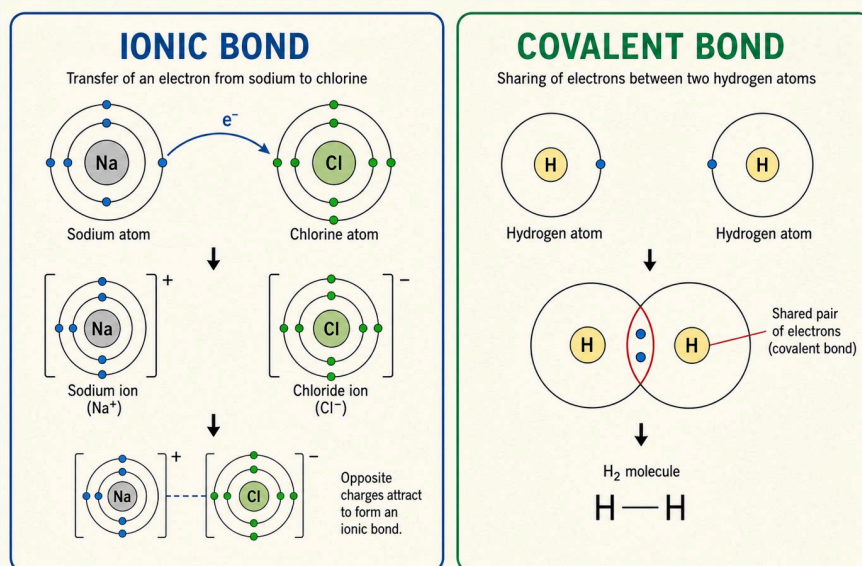


Figure 2.4 An ionic bond transfers an electron; a covalent bond shares a pair.

Covalent molecules: oxygen, carbon dioxide and ammonia

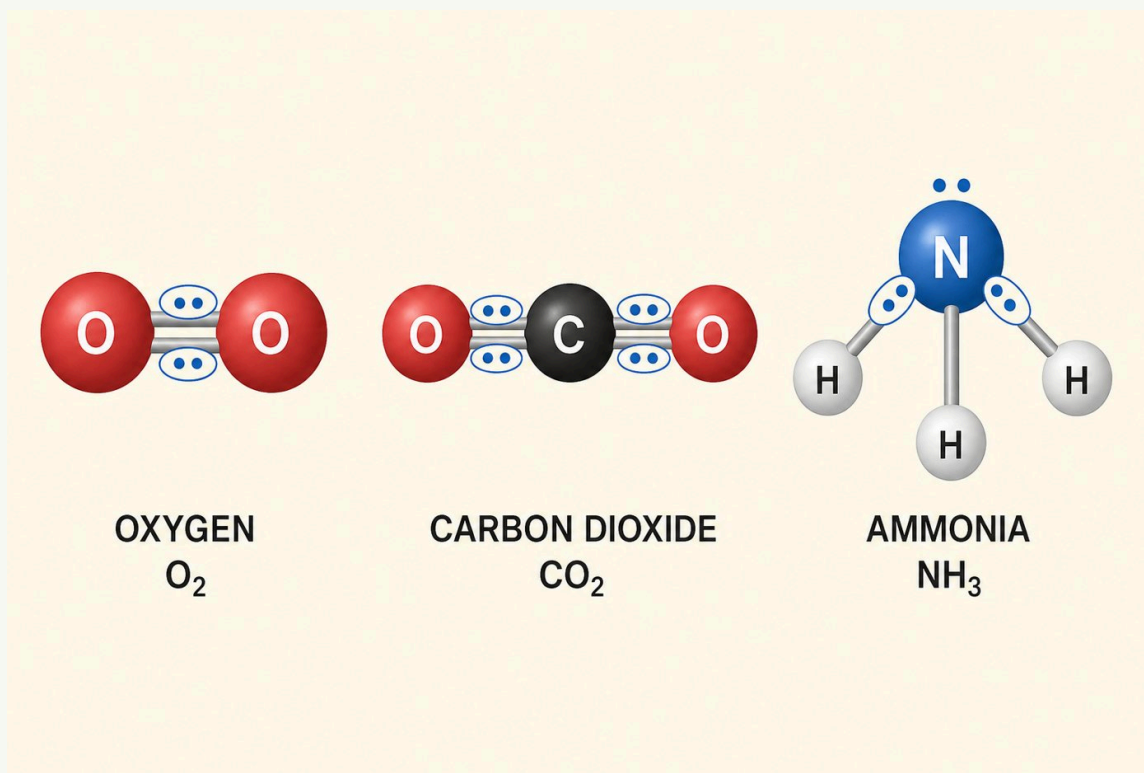


Figure 2.3 Three covalent molecules, with their shared electron pairs circled.

Oxygen (O_2) is simply two oxygen atoms sharing two pairs of electrons, a double bond. Each atom needed two more electrons to fill its shell, and both are satisfied by the same two shared pairs. This is the molecule you breathe; nothing else shares its electrons so generously.

Carbon dioxide (CO_2) is a straight line of three atoms: oxygen, carbon, oxygen again, each pair joined by a double bond. Notice how one carbon atom completes its shell by sharing with two neighbours at once. Plants take in this molecule and split it apart to build sugar.

Ammonia (NH_3) is a pyramid: one nitrogen atom on top, three hydrogens around its base, each joined by a single shared pair. Nitrogen came to covalent bonding needing three electrons, so it makes three bonds. That spare pair of electrons sitting alone on top of the nitrogen is what gives ammonia its sharp smell and its use in cleaning products.

More covalent molecules: chlorine, water and methane

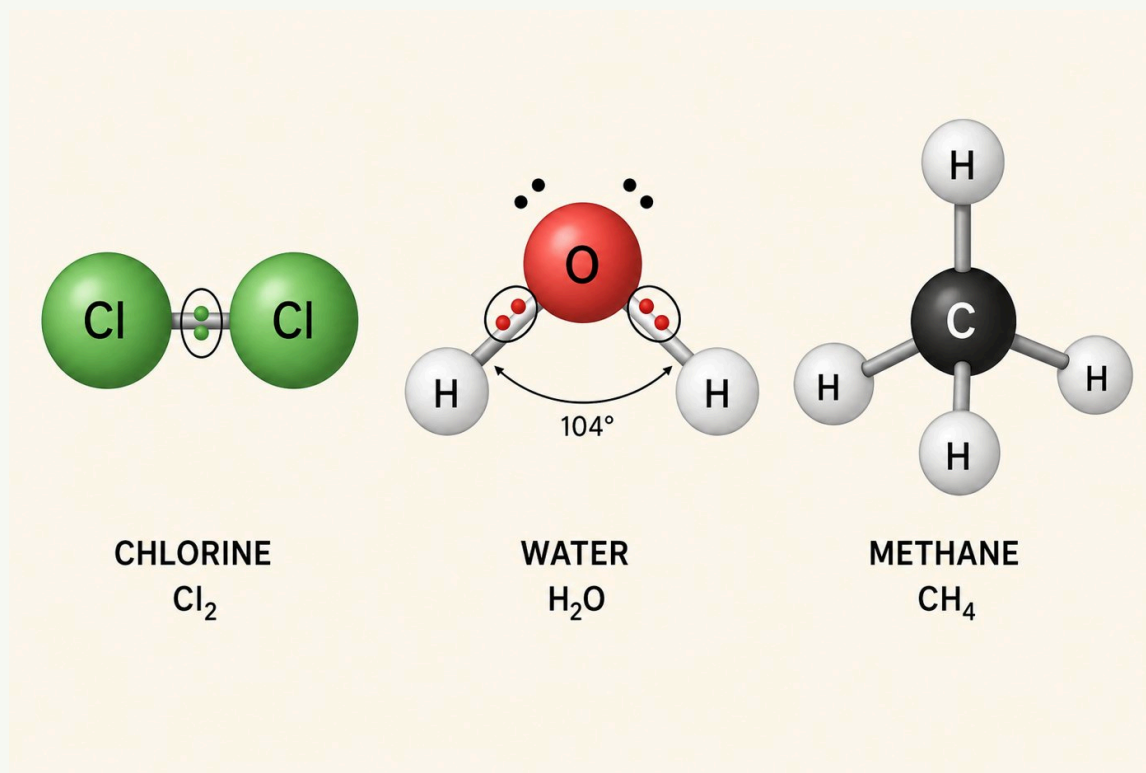


Figure 2.4 Three more covalent molecules: chlorine, water and methane.

Chlorine (Cl_2) is the simplest picture of covalent bonding: two identical atoms, each needing one more electron, sharing a single pair between them. Because both atoms pull equally, the pair sits exactly halfway. This pale green gas is the same chlorine that purifies swimming pools, and it is so eager to complete its shell that it reacts with almost anything that can give an electron away.

Water (H_2O) is not straight, but bent. Oxygen needed two more electrons, so it makes two bonds, one to each hydrogen, with two lone pairs left over sitting on top. Those lone pairs push the bonds closer together, squeezing the angle to about 104° . That bent shape, with its uneven sharing, is why water dissolves salt, climbs up plant stems and freezes into a lattice that floats.

Methane (CH_4) is carbon's tidy masterpiece. Carbon needed four electrons, so it makes four identical bonds to four hydrogens, arranged as a three-pointed pyramid in three dimensions, called a tetrahedron. Each bond points as far from the others as geometry allows. Methane is the main gas in natural gas, and the same four-bonded carbon builds everything from petrol to proteins.

Metallic bonding: the third kind

A metal bond is a bond to the crowd. Ionic bonding transfers electrons from one atom to another; covalent bonding shares them between two. Metallic bonding does neither. Instead, every atom in the metal donates its loose outer electrons to the whole structure at once. The atoms left behind become positive ions, packed in regular rows, while the donated electrons drift freely between them, a wandering swarm called a sea of electrons.

The glue is attraction. What holds the metal together is the pull between the positive ions and the negative sea around them. The electrons belong to no single atom, so the bonding spreads through the entire lump of metal in every direction, which is why metals are strong yet flexible rather than brittle like salt crystals.

The sea explains everything. Because the free electrons drift wherever a push sends them, metals conduct electricity: connect a battery and the whole sea flows. The packed ions let whole layers slide past each other without breaking the bond, which is why metals bend, hammer flat and draw into wires. And the moving electrons carry energy swiftly from hot end to cold, which is why metals feel cold to the touch and make such good pans.

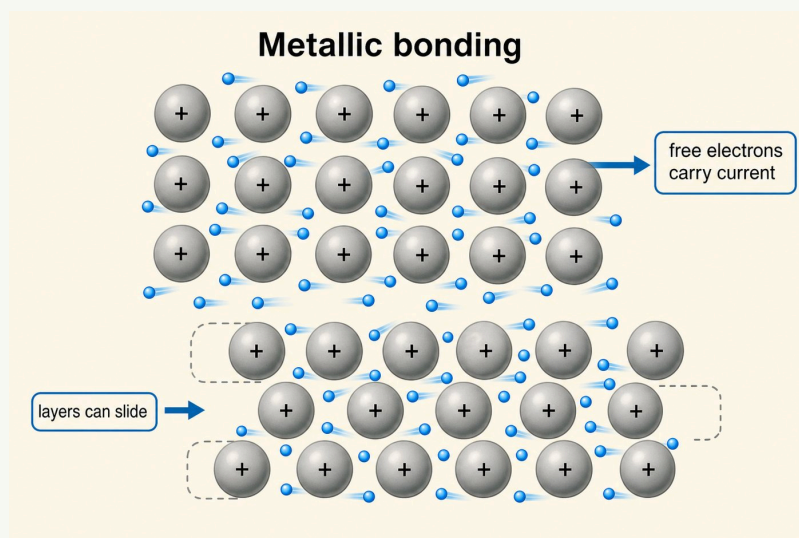


Figure 2.5 Metallic bonding: positive ions in a sea of free electrons.

THE THREE BONDS AT A GLANCE

Ionic: electrons transferred, ions formed, metal with non-metal. **Covalent:** electrons shared between two non-metals. **Metallic:** electrons donated to a shared sea, metal with metal.

Activity: the three types of bonding

Work through the tasks in order. They build from sorting, to modelling, to explaining.

· ACTIVITY

Sort the bonds. For each substance, decide whether it is held together by ionic bonds, covalent bonds or metallic bonds, and write a one-line reason: sodium chloride; water; copper; methane; magnesium oxide; oxygen gas. Two are ionic, three are covalent and one is metallic. Check your reasons mention transferring, sharing or a sea of electrons.

Build the bonds. In pairs, use modelling clay or beads. Build one ionic pair (a sodium ion that has lost an electron and a chloride ion that has gained it, showing the charges), one covalent molecule (water, with shared pairs circled), and one metallic lattice (positive ions in a grid with loose electrons scattered between them). Label every particle.

Swap and spot. Exchange models with another pair. Identify which type of bonding each model shows, and point to the single feature that gives it away. A model with charges shows transfer; overlapping shells show sharing; free-floating electrons show a metal.

Predict and explain. Use your models to predict: which substance will have the highest melting point, sodium chloride or water? Which will conduct electricity when solid, copper or sodium chloride, and which only when dissolved or molten? Write two sentences explaining each answer using the bonding you modelled.

Think like a scientist. A pupil says, 'Water must be ionic because it is made of two different elements.' Write three sentences correcting them: what actually decides the bonding type, why water is covalent, and one test you could use to tell them apart in the laboratory.

Simple and giant structures

Learn

How the particles in a substance are arranged explains how it behaves. In **simple molecular substances** such as water, carbon dioxide and methane, a few atoms are held tightly together in molecules, but the molecules attract each other only weakly. Little energy is needed to pull them apart, so these substances melt and boil at low temperatures and many are gases or liquids at room temperature.

Metals are different again. Their atoms pack in a **giant metallic structure** of positive ions in a sea of loose outer electrons. The electrons wander freely, which is why metals conduct electricity so well; the packed ions let layers slide, which is why metals can be bent and hammered into shape.

Ionic compounds such as sodium chloride form a **giant ionic lattice**, a repeating pattern of alternating positive and negative ions held by strong attraction. It takes great energy to break, so ionic compounds have high melting points, and they conduct electricity only when dissolved or molten, when their ions are free to move.

· QUESTIONS

- 1 Explain why carbon dioxide is a gas at room temperature while sodium chloride is a solid with a high melting point.
- 2 Use the giant structure of metals to explain why copper is used for electrical wiring.
- 3 Solid sodium chloride does not conduct electricity, but the molten substance does. Explain why.

Diamond and graphite: two giant covalent structures

Diamond and graphite are both made of nothing but carbon atoms, yet they could not be more different. The secret is the bonding. In diamond, every carbon atom uses all four of its bonds, reaching out to four neighbours in a rigid three-dimensional network that repeats without end. Nothing in the structure can move or slip, so diamond is the hardest natural material known, and its tightly held electrons give it a sparkling, glassy brilliance.

Graphite bonds differently. Each carbon atom joins only three neighbours, locking atoms into flat sheets of hexagons, stacked layer upon layer. The fourth electron of every atom is left free to wander, which is why graphite conducts electricity, alone among non-metals. The layers themselves attract each other only feebly, so they glide apart at the lightest touch: that is what happens when a pencil slides across paper, leaving a smear of layers behind.

Same element, opposite characters. One is a transparent gem used to cut glass; the other is a black, slippery solid that lubricates locks and powers electrodes. Bonding, not the kind of atom, decides everything.

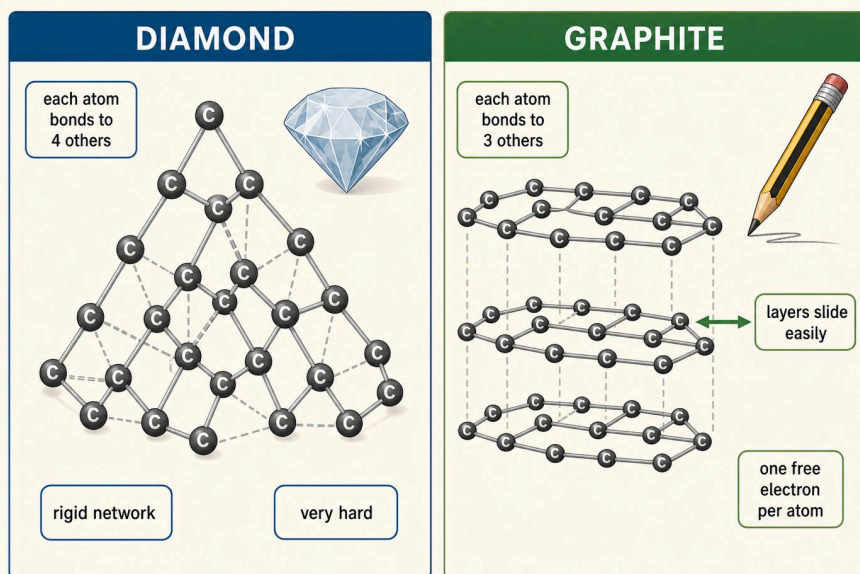


Figure 2.5 Diamond's rigid four-bond network against graphite's sliding three-bond layers.

GLOSSARY

Unit 2 · Properties of materials

atom: the smallest particle of an element that still has the properties of that element.

proton: a positively charged particle inside the nucleus.

neutron: a particle in the nucleus with no charge.

electron: a negatively charged particle moving in shells around the nucleus.

atomic number: the number of protons in an atom; it fixes the element.

mass number: the total number of protons and neutrons in an atom.

Periodic Table: the chart of elements arranged by atomic number into periods and groups.

compound: two or more elements chemically joined together.

covalent bond: a shared pair of electrons between two non-metal atoms.

ion: an atom that has lost or gained electrons and so carries a charge.

giant structure: a lattice of millions of particles bonded together, as in metals and ionic solids.

3

Forces and energy

Why things sink or float,
and where energy really goes

- IN THIS UNIT, YOU WILL
- Define and calculate density
- Distinguish heat from temperature
- Apply conservation of energy
- Describe how thermal energy moves from hot to cold
- Explain cooling by evaporation

Density

Learn

Which is heavier, a kilogram of steel or a kilogram of feathers? Neither, of course; they simply pack their mass differently. **Density** describes how tightly packed a substance is:

$$\text{density} = \text{mass} \div \text{volume}$$

Density is measured in grams per cubic centimetre (g/cm^3) or kilograms per cubic metre (kg/m^3). Water has a density of 1.0 g/cm^3 , and this makes a convenient benchmark: objects less dense than water float, and objects denser than water sink. Steel is about 8 g/cm^3 and sinks; pinewood is about 0.5 g/cm^3 and floats.

You can find the density of a regular block by measuring its sides with a ruler and its mass on a balance. For an irregular object, such as a pebble, measure the volume by **displacement**: lower it into a measuring cylinder of water and read how far the water level rises.

• QUESTIONS

- 1 A block has a mass of 60 g and measures $2 \text{ cm} \times 3 \text{ cm} \times 5 \text{ cm}$. Calculate its density in g/cm^3 .
- 2 A stone of mass 96 g displaces 32 cm^3 of water. Find its density and predict whether it floats in water.
- 3 Explain why a ship made of steel floats, even though steel is denser than water.

Finding the density of regular and irregular objects

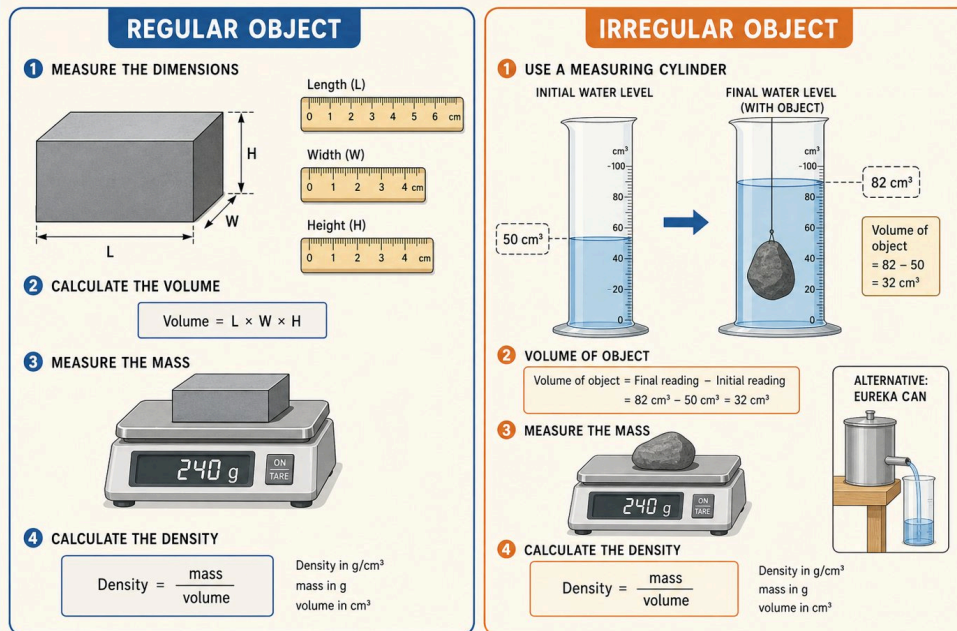


Figure 3.2 Measuring density: a regular block by ruler, an irregular stone by displacement.

• METHOD

Regular object. Measure the length, width and height of the block with a ruler to the nearest 0.1 cm, and multiply them to find the volume in cm³. Place the block on a balance to find its mass in grams, checking the balance reads 0.0 g first. Divide mass by volume to find the density.

Irregular object. Half-fill a measuring cylinder with water and record the starting volume. Lower the stone gently on a thread until it is fully submerged, and record the new volume. The rise in the water level equals the stone's volume. Weigh the stone, then divide mass by volume.

Make it fair. Repeat each measurement three times and use the mean. Read the cylinder at eye level, with your eye level with the bottom of the meniscus, and record results in a table with units.

Sink or float?

Density at work. In the tank, wood and ice float because their particles are more spread out than water's, so each cubic centimetre weighs less than a cubic centimetre of water; steel sinks because its particles are packed far more tightly. The arrows show upthrust, the upward push of the water, which only equals the weight of a floating object when that object is less dense than the water. Notice the balance too: the small steel cube and the large foam block weigh exactly the same, showing that a big object can still be light if its mass is thinly spread. Size never decides floating; density does.

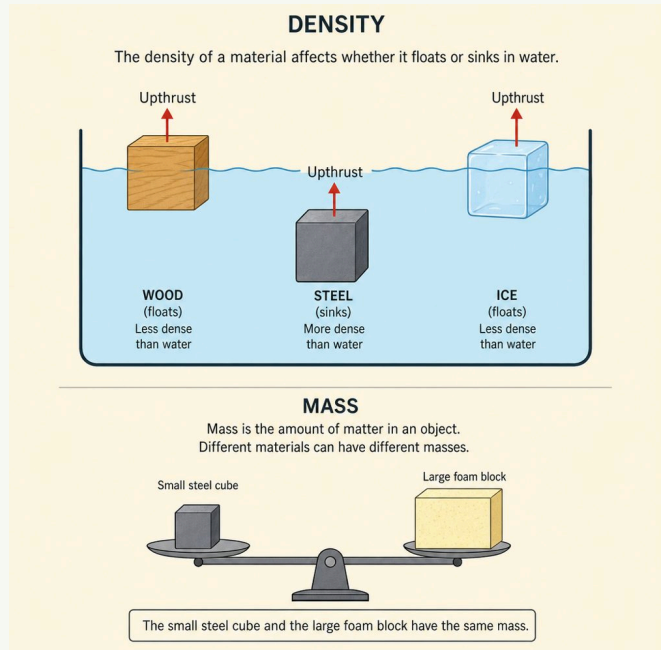


Figure 3.1

• ACTIVITY

Calculate and predict. Calculate the density of each object, then predict whether it floats in water (density 1.0 g/cm^3). Object A: mass 24 g, volume 12 cm^3 . Object B: mass 90 g, volume 100 cm^3 . Object C: mass 480 g, volume 40 cm^3 . Explain each prediction.

Kitchen investigation. With permission, test three waterproof household objects in a bowl of water. Record your predictions before testing, then explain any surprises using the idea of density.

Think like a scientist. A ship is made of steel, which is about 8 g/cm^3 , yet the ship floats. Draw a labelled diagram to explain this apparent paradox.

Heat and temperature

Learn

Heat and temperature are not the same thing, and confusing them is the most common mistake in this topic. **Temperature** tells you how fast the particles in a substance are moving on average. **Heat** is thermal energy on the move, flowing always from a hotter place to a colder one.

A bathtub of lukewarm water holds far more thermal energy than a red-hot nail, because there are so many more particles sharing the energy between them. A cup of tea at 80 °C will not warm a swimming pool; a bath at 40 °C will warm you, because it has the bulk to keep giving.

Temperature is measured with a thermometer in degrees Celsius. Heat is measured in joules (J), like all energy. Keep the two ideas separate in your mind: temperature is the reading on the dial, heat is the flow of energy down the gradient.

· QUESTIONS

- 1 A red-hot nail is dropped into a bucket of cool water. In which direction does heat flow, and when does it stop?
- 2 A cup of tea and a bath both contain water at 60 °C and 40 °C respectively. Explain which one holds more thermal energy.
- 3 State the units of temperature and of heat.

Heat and temperature: which holds more energy?

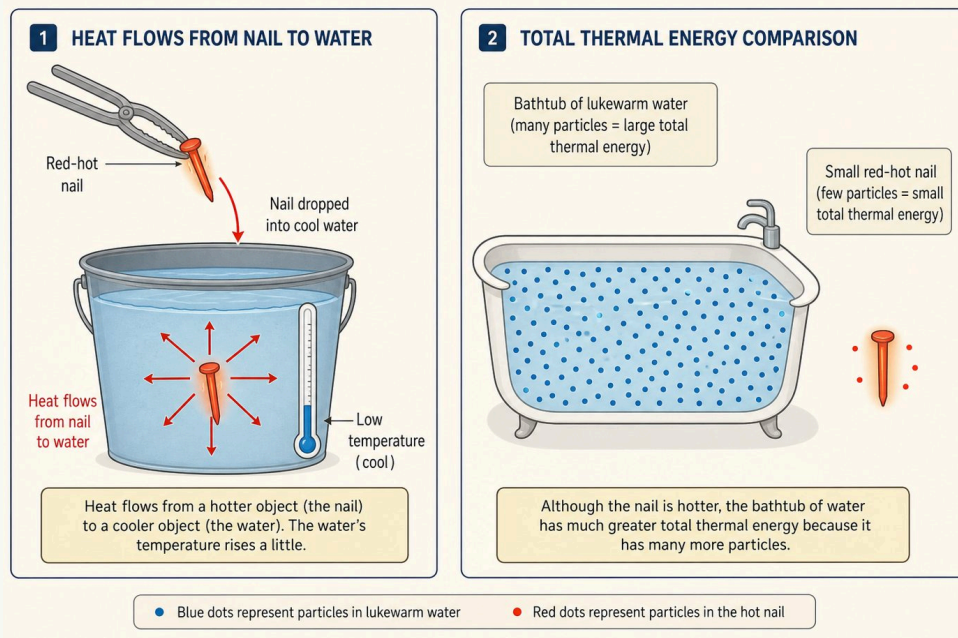


Figure 3.3 Heat always flows from hotter to cooler; total energy depends on particle numbers, not temperature.

• METHOD

You will need. A large beaker or bucket of cool water, a warm metal spoon or safely heated small metal object handled with tongs, two thermometers, a stopwatch.

Part 1: which way does heat flow? Record the water temperature. Lower the hot object into the water and stir gently. Record the temperature every minute for five minutes. The water warms and the object cools, until both reach the same temperature and the readings stop changing. Heat has flowed from hotter to colder, and never the other way.

Part 2: temperature is not energy. Compare two thermometer readings: a small cup of water at 80 °C and a bucket at 40 °C. Predict which could warm a cold room more. Test by touching the outside of each container (carefully). The bucket holds far more thermal energy because it has so many more particles, even though its temperature is lower.

Make it fair. Use the same amount of water in Part 1 for every trial, keep the thermometer bulbs fully submerged but not touching the container, and read each thermometer at eye level.

Conservation of energy

Learn

One of the grandest rules in science says that energy can neither be created nor destroyed, only transferred from one store to another. This is the **conservation of energy**.

Energy is held in stores: kinetic, gravitational potential, chemical, elastic, thermal and others. A dropped ball trades gravitational potential energy for kinetic energy as it falls, then, on landing, passes almost all of it to the thermal store of the ground and the air. Nothing has vanished; it has simply spread out where it is harder to use.

Energy that spreads into the surroundings like this is called **dissipated** energy. It is not lost, but it is no longer concentrated enough to be useful. Every machine, every living thing and every power station obeys the same rule, which is why engineers care so much about reducing unwanted transfers.

· QUESTIONS

- 1 Describe the energy transfers when a ball is dropped from your hand until it comes to rest on the floor.
- 2 What is meant by dissipated energy, and why is it not the same as destroyed energy?
- 3 Explain why a torch obeying conservation of energy eventually goes dark.

Moving from hot to cold

Learn

Thermal energy always transfers from hotter places to colder places, never the other way round. Your warm hands and a cold drink can only end up at one temperature: the point between their two starting temperatures.

This one-way traffic explains everyday puzzles. When a window is opened on a cold day, thermal energy flows out of the room, and the room feels colder. A metal spoon in hot soup feels hot quickly because metal conducts the energy towards your hand; a wooden spoon hardly warms at all, because wood is a poor conductor, a thermal **insulator**.

Transferring thermal energy is called **heat dissipation** when the energy spreads into stores where it was not wanted. The surroundings are always the final destination.

· QUESTIONS

- 1 Explain, in terms of thermal energy, why a room cools when a window is opened on a cold day.
- 2 Why does a metal spoon feel colder than a plastic one at the same room temperature?
- 3 Describe the eventual fate of the thermal energy from a hot cup of coffee left on a desk.

Ways of transferring thermal energy

Learn

Thermal energy travels in three ways. **Conduction** is the passing of energy from particle to particle, mainly in solids; metals conduct best because their free electrons carry energy swiftly. **Convection** happens in liquids and gases: a warm fluid expands, becomes less dense, rises, and cooler fluid sinks to take its place, setting up a **convection current** that circulates the energy. **Radiation** needs no particles at all; every object emits infrared radiation, which is how the Sun's energy crosses empty space to reach us.

Houses are designed around these three roads. Loft insulation traps air, which is a poor conductor; draught excluders block convection loops; shiny foil behind radiators reflects radiation back into the room.

· QUESTIONS

- 1 Name the main method of thermal transfer in a solid metal rod, in boiling water, and from the Sun to the Earth.
- 2 Explain how a convection current forms in a pan of water being heated from below.
- 3 Give one design feature of a house that reduces each of the three methods of transfer.

Cooling by evaporation

Learn

After swimming, you shiver even on a warm day. The water on your skin is **evaporating**: its fastest-moving particles escape into the air as vapour. Only the speediest particles can break free, so when they leave, the particles left behind are the slower, cooler ones. Your skin loses energy and you feel cold.

Evaporation is therefore a cooling process. It happens at any temperature, unlike boiling, but it speeds up when the surface is warm, when the surface area is large, and when the air above carries the vapour away, as it does on a windy day.

Living things exploit this. Sweating cools you because evaporating water carries thermal energy away from your skin; dogs pant to evaporate water from their tongues; even a wet clay pot keeps its contents cool in hot climates by the same trick.

· QUESTIONS

- 1 Explain, in terms of particles, why evaporation cools the liquid left behind.
- 2 Give three factors that increase the rate of evaporation.
- 3 Why does sweating fail to cool you well on a hot, humid day?

Heat on the move

Three roads for thermal energy. From the Sun, radiation travels as infrared waves, crossing empty space with no particles at all. In the pan on the hob, conduction passes energy along the metal from particle to particle, which is why the handle grows warm even though only the base touches the heat; metals conduct best because their loose electrons carry the energy swiftly. Around the heater, convection carries warmth through the air: hot air expands, becomes less dense and rises, and cooler air sinks to take its place, setting up the circulating current shown by the curved arrows. Use the legend at the foot of the diagram to tell the straight heat-flow arrows from the wavy radiation and the looping convection currents.

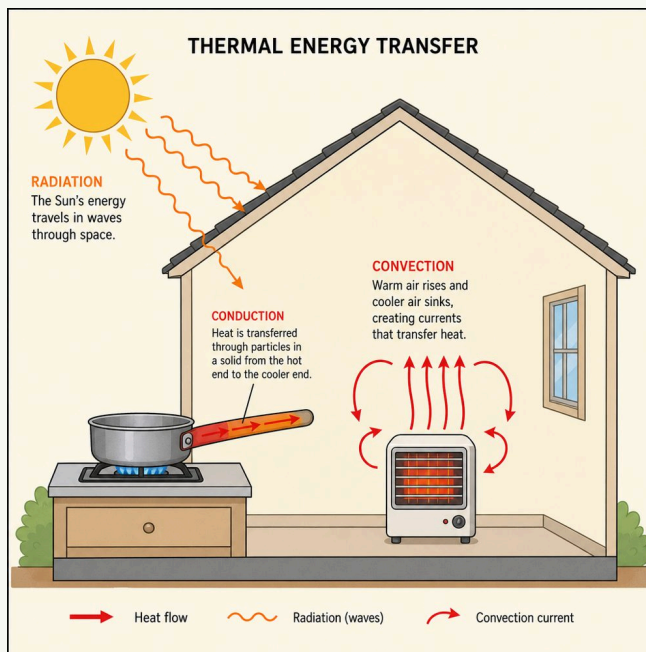


Figure 3.4

· ACTIVITY

Name the method. For each situation, name the main method of thermal transfer and explain your choice: a metal spoon heating up in soup; the warm air above a radiator drifting to the ceiling; feeling the warmth of a bonfire from two metres away; the handle of a bicycle frame going cold in winter.

Design a heat-proof house. Draw a cross-section of a house and add three features that reduce energy loss, one for each method of transfer. Annotate each feature with the method it blocks and how it works.

Convection in colour. With supervision, drop a crystal of potassium manganate(VII) or a little food colouring into a beaker of cold water and warm it gently from below. Sketch and explain the currents you see.

GLOSSARY

Unit 3 · Forces and energy

density: mass per unit volume, calculated as $\text{mass} \div \text{volume}$.

displacement method: finding an object's volume from the rise in water level it causes.

temperature: a measure of the average speed of the particles in a substance, in °C.

heat: thermal energy transferred from a hotter place to a colder one, measured in joules.

conservation of energy: the rule that energy can be transferred but never created or destroyed.

dissipated energy: energy spread into the surroundings, still present but no longer useful.

conduction: transfer of thermal energy from particle to particle, mainly in solids.

convection: transfer of thermal energy by circulating currents in a liquid or gas.

radiation: transfer of energy by infrared waves, needing no particles.

evaporation: escape of the fastest particles from a liquid surface, cooling what remains.

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